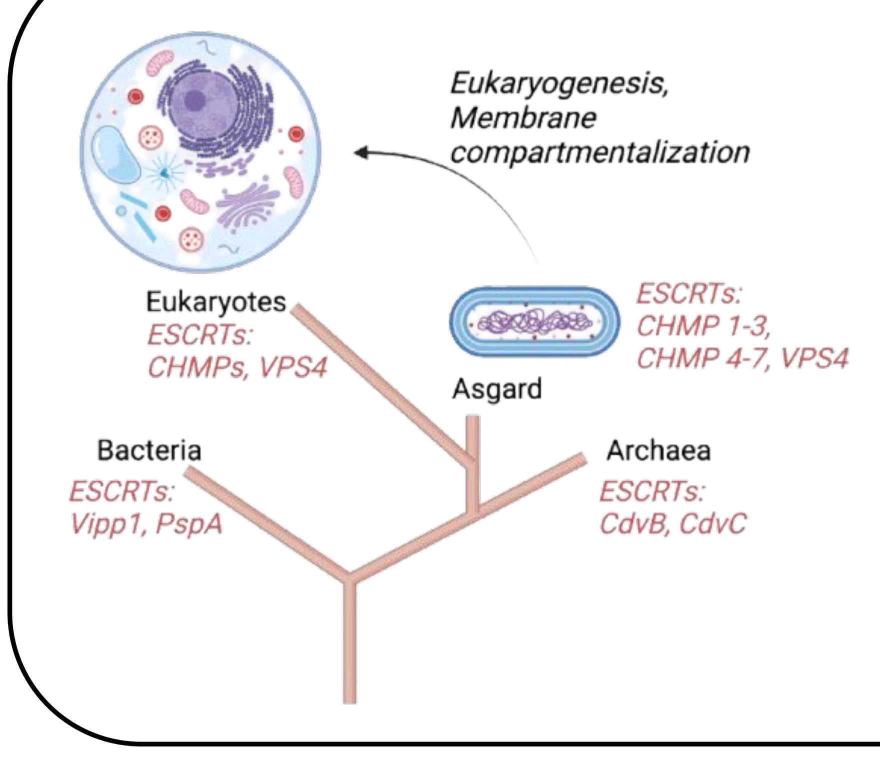
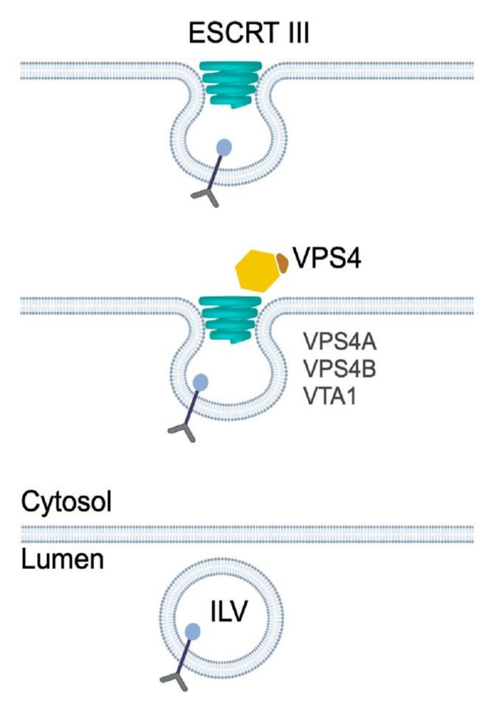
Environment drives residue-level adaptations while phylogeny dictates the biochemical backbone of ESCRT proteins in Asgard archaea

Noam Guetta [1,2] (guettano@post.bgu.ac.il), Prof. Shai Pilosof[1,3], Prof. Natalie Elia[1]

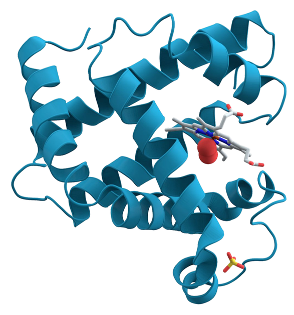
1. Department of Life Sciences, Ben-Gurion University of the Negev. 2. Jusidman Science Center for Youth, Ben Gurion University of the Negev, Beer Sheva, Israel & Alpha program, Future Scientists Program, Tel Aviv, Israel. 3. The Goldman Sonnenfeldt School of Sustainability and Climate Change, Ben-Gurion University of the Negev.

Introduction & hypothesis

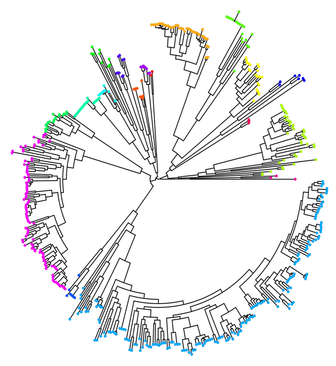




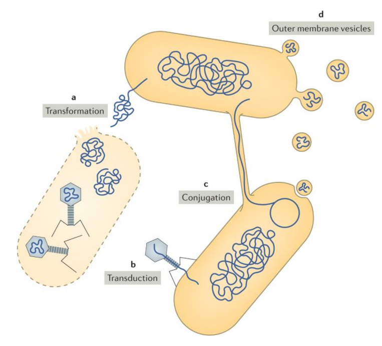
Similarity ~ Phylogeny + Geography + Environment + ε



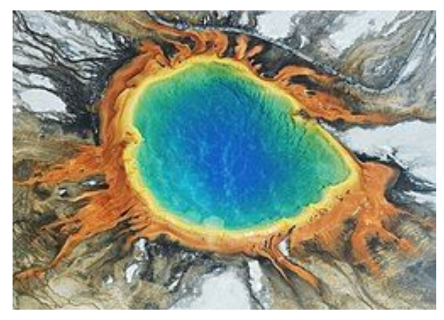
Sequence, Structure



Mutations, Evolution, ...

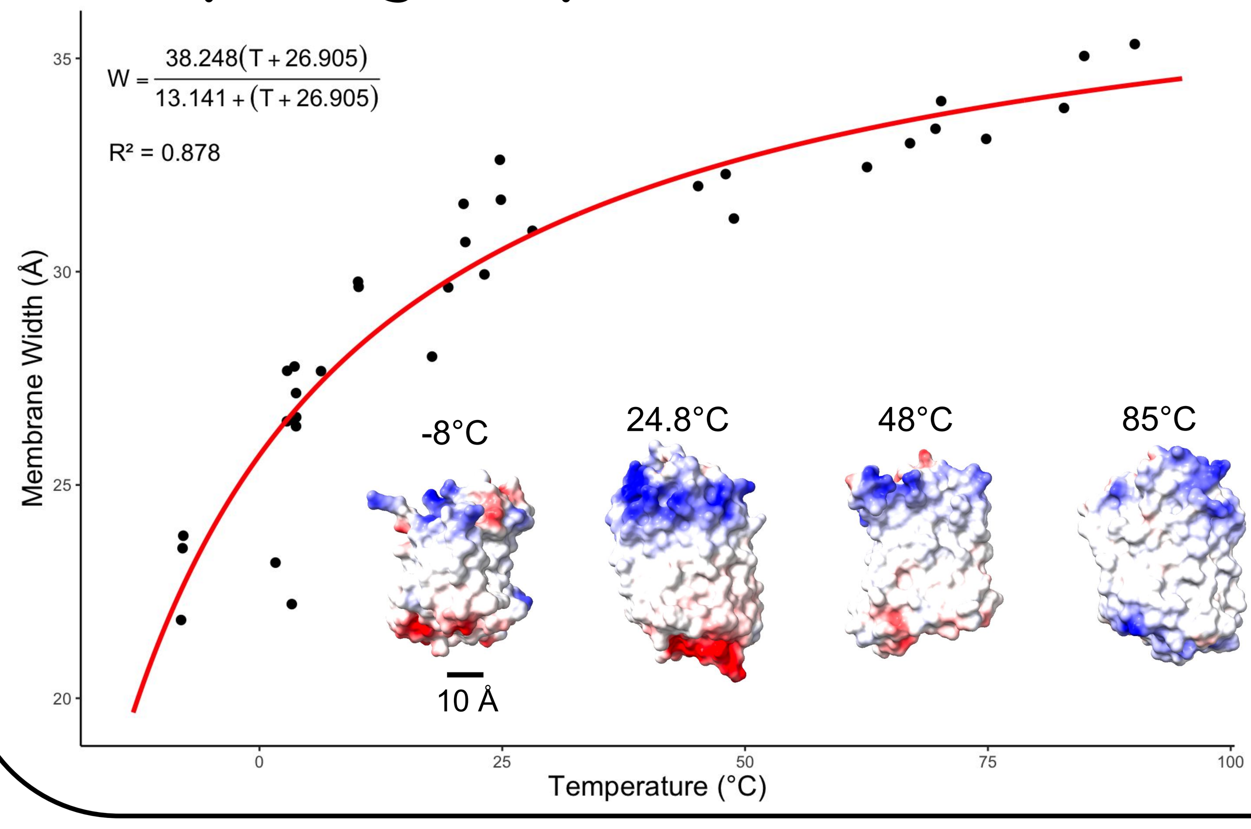


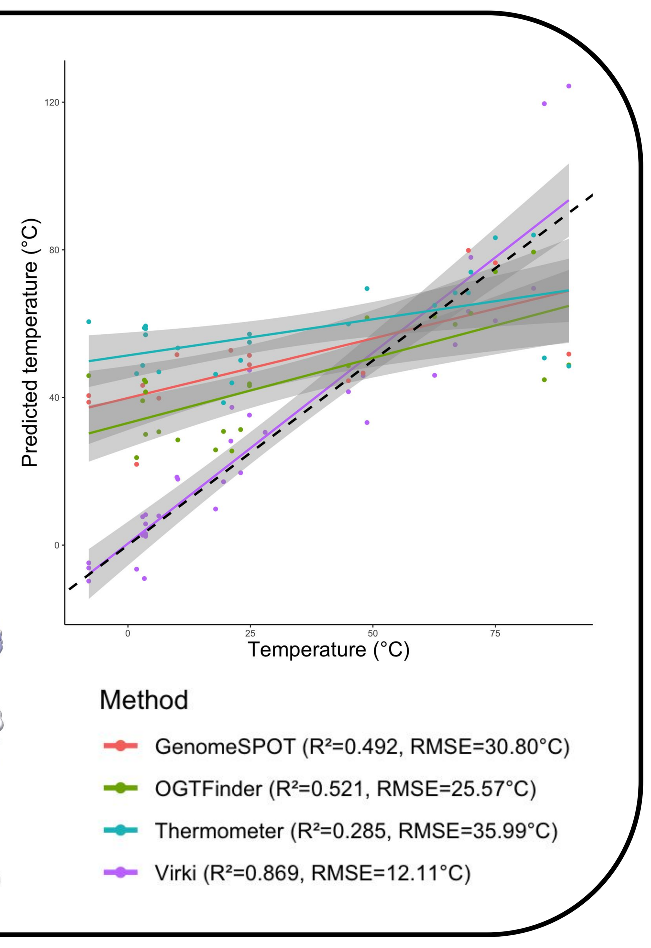
Movement, HGT, ...



Temperature, Salinity, ...

Completing temperature data





Measuring protein distance with Bray-Curtis

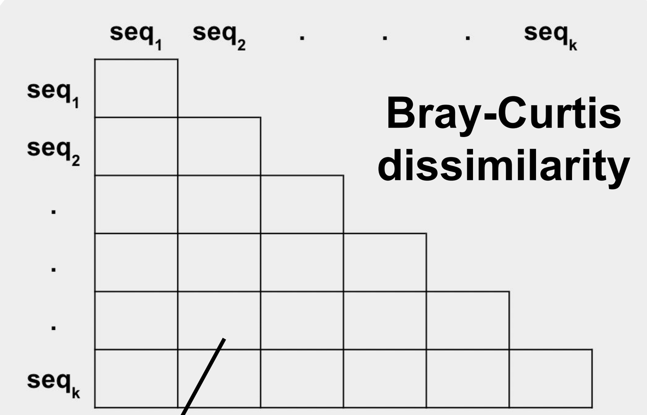
Seq_1: -------...IKDEIAESELK----------

Seq_2: -----------...LTDELKEGDT--------

...

Seq_n: ---------...LQDEVEKKELKS-----

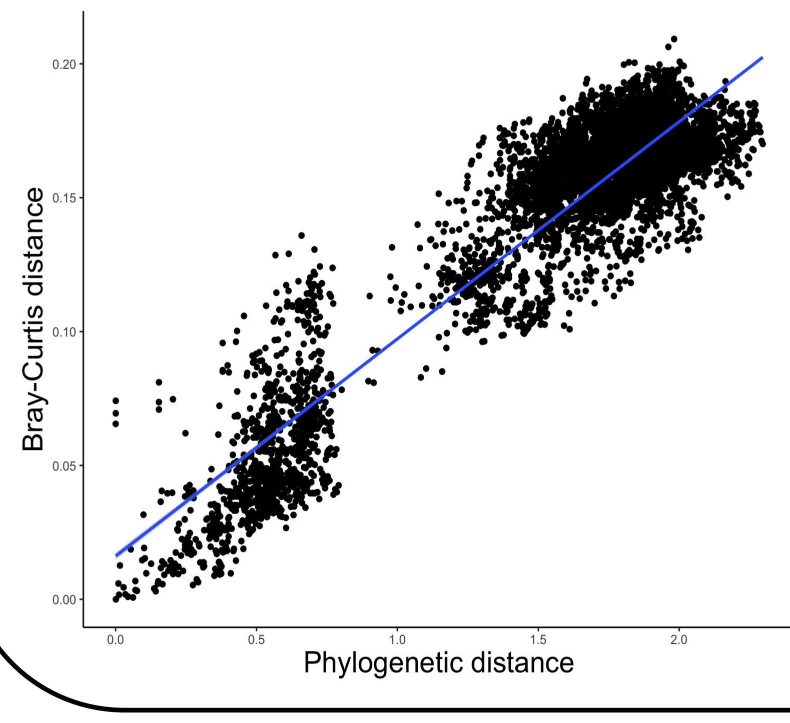
v_x = ... (Hyd_{93}, $Polar_{93}$, $Bulk_{93}$), (Hyd_{94}, $Polar_{94}$, $Bulk_{94}$) ...

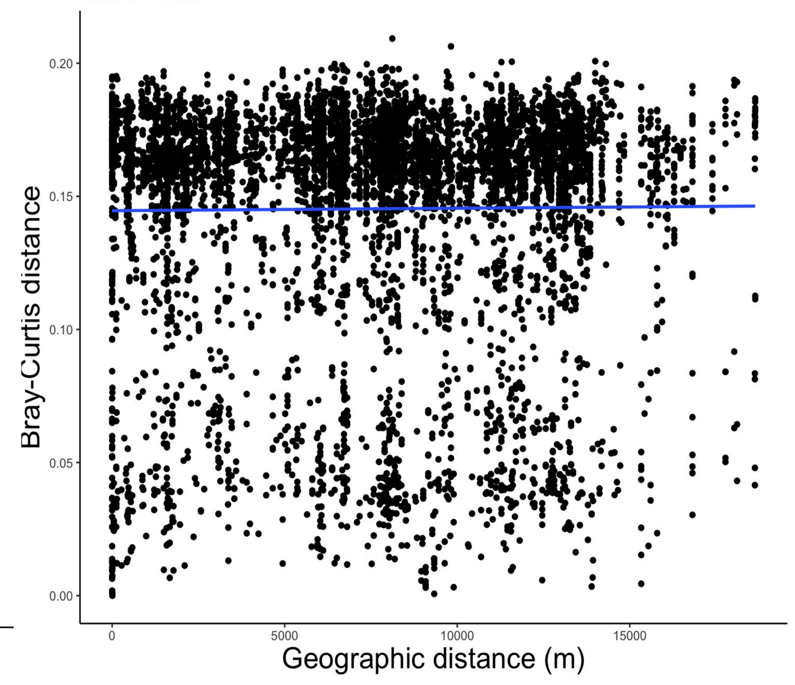


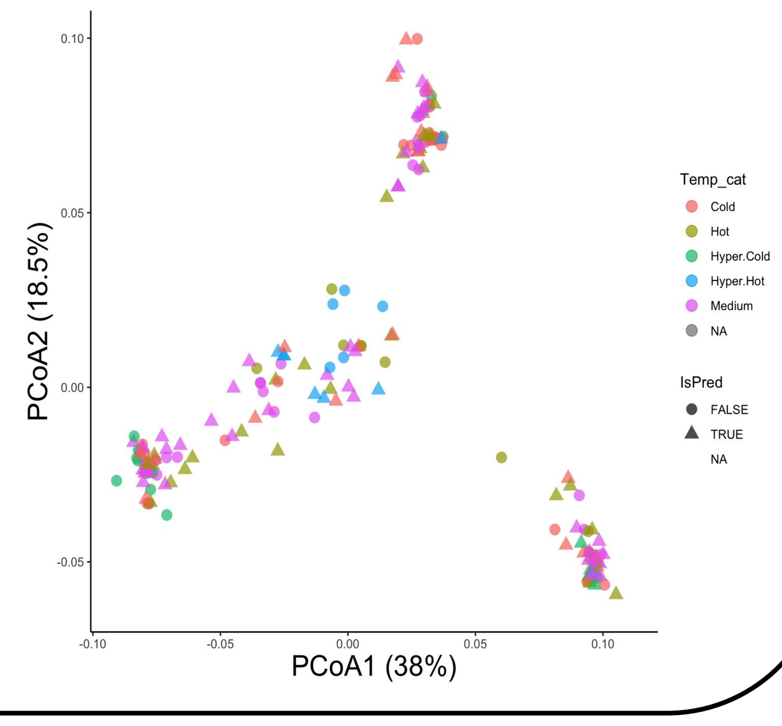
$$D_{(x,y)} = \frac{\sum_{i=1}^{3n} |x_i - y_i|}{\sum_{i=1}^{3n} x_i + y_i}$$

Phylogeny is the main driver of sequence similarity

Phylogeny + Geography + Environment







Finding residue-level changes

For each residue

v_1: (0.00, 0.57, 0.62 ... 0.46, 0.21)

v_2: (0.69, 0.12, 0.29, ... 0.05, 0.48)

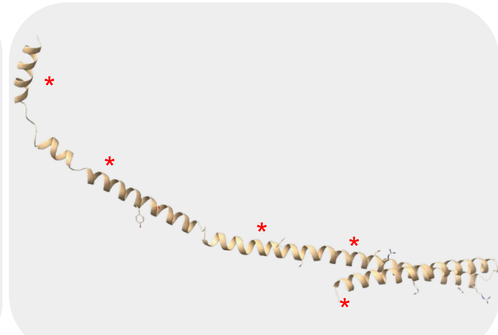
...

v_n: (0.78, 0.56, 0.67, ... 0.73, 0.90)

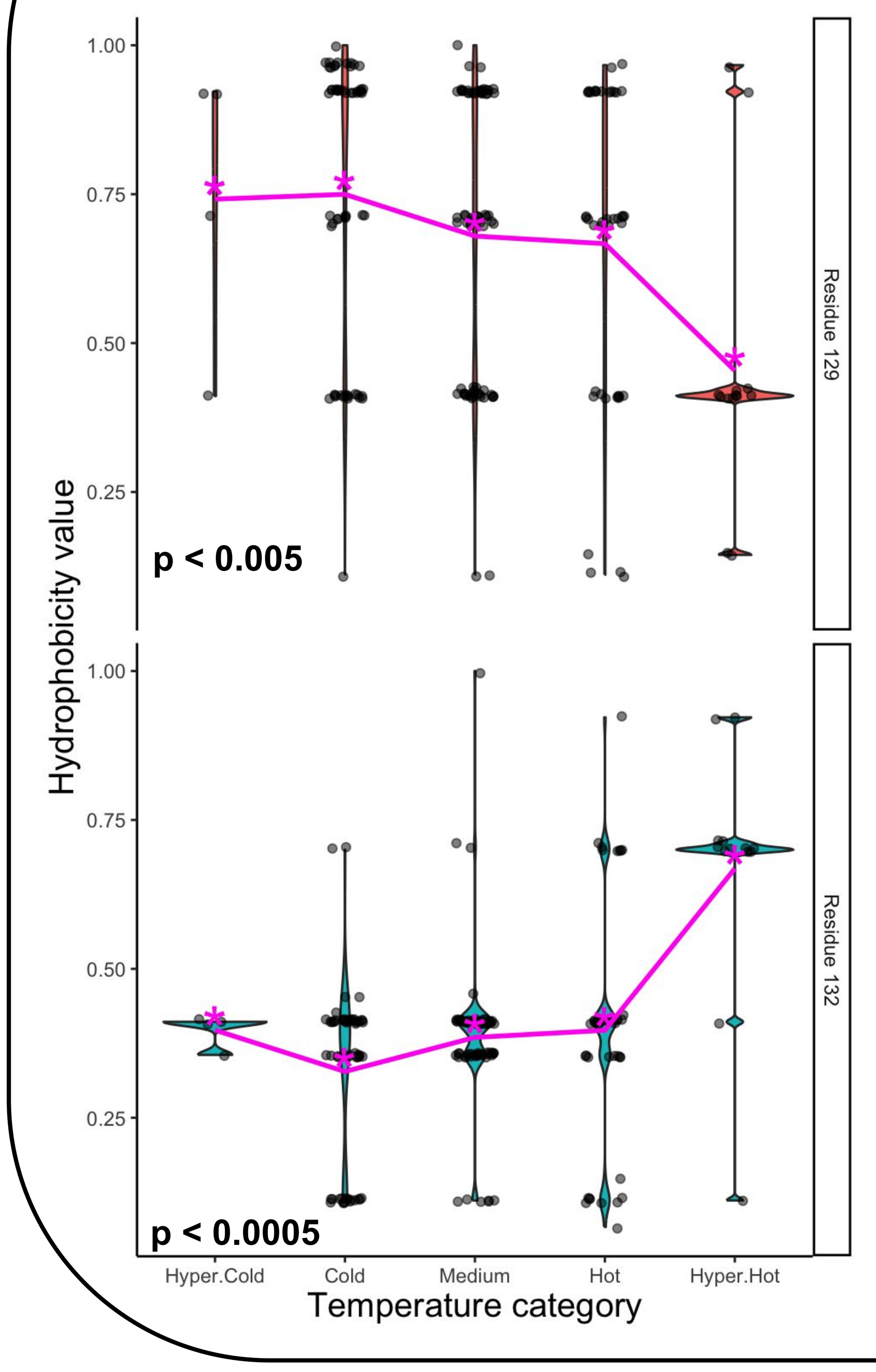
ANOVA (p) + Pearson (r)

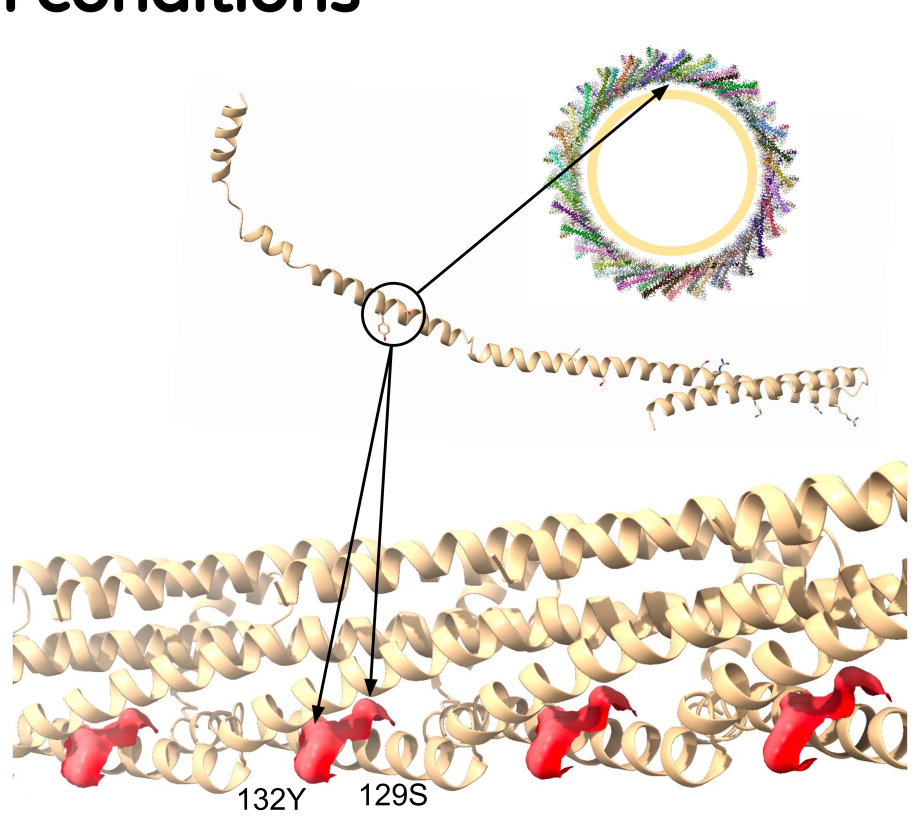
Results list

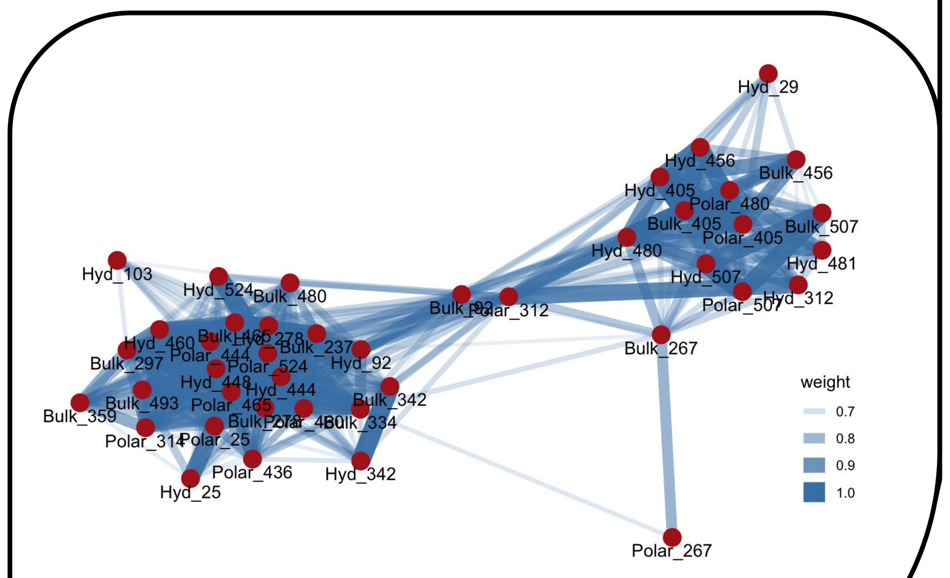
Residue	p_{adj}	r
Hyd_{101}	0.003	0.53
$Polar_{291}$	0.192	0.04
...	...	...



Residue-level changes and interactions are induced by environmental conditions







Correlation network reveals changing interactions

Conclusions

* Using biochemical properties reveals changes in sites with predicted functions and interactions.
* Phylogeny is the main driver of the biochemical backbone and geography shows no strong effect.
* Environment induces residue-level changes with predicted functionality.

Bibliography

1. Zaremba-Niedzwiedzka, K. et al. Asgard archaea illuminate the origin of eukaryotic cellular complexity (2017).
2. Nachmias, D. et al. ESCRTs – a multi-purpose membrane remodeling device encoded in all life forms (2025).
3. Panagiotou, K., Geesink, P., Köstlbacher, S. et al. Diversity, ecology, cell biology and evolution of the Asgard archaea (2026).

Want to know more?

Look at the online poster at our website!